

Lecture 4: Numerically Solve RBC Model

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Solve LOM

$$\hat{k}_t = \frac{1}{\beta} \hat{k}_{t-1} + \frac{\tilde{\delta}}{\alpha\beta} \hat{z}_t - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) \hat{c}_t$$

$$\sigma E_t \hat{c}_{t+1} + \tilde{\delta}(1 - \alpha) \hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1} = \sigma \hat{c}_t$$

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t$$

State variables: $\hat{k}_{t-1}$ and shock $\hat{z}_t$.

$\hat{k}_{t-1}$ is predetermined.

The dynamics of the model can be described as a recursive law of motion in terms of states.

LOM

$$\begin{cases} \hat{k}_t = V_{kk} \hat{k}_{t-1} + V_{kz} \hat{z}_t, \\ \hat{c}_t = V_{ck} \hat{k}_{t-1} + V_{cz} \hat{z}_t, \end{cases}$$

and solve for $(V_{kk}, V_{kz}, V_{ck}, V_{cz})$ by the method of undetermined coefficients.

1)

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t \quad \Rightarrow \quad E_t \hat{z}_{t+1} = \psi \hat{z}_t.$$

Since $E_t(\varepsilon_t) = 0$, and $\varepsilon_t \sim iid N(0, \sigma^2)$,

$$E_t(\hat{z}_t) = \hat{z}_t,$$

because Z_t is already realized in period t , hence belongs to period- t 's information set.

2) From

$$\hat{k}_t = \frac{1}{\beta} \hat{k}_{t-1} + \frac{\tilde{\delta}}{\alpha\beta} \hat{z}_t - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) \hat{c}_t,$$

substitute

$$\hat{k}_t = V_{kk} \hat{k}_{t-1} + V_{kz} \hat{z}_t, \quad \hat{c}_t = V_{ck} \hat{k}_{t-1} + V_{cz} \hat{z}_t,$$

to get

$$V_{kk} \hat{k}_{t-1} + V_{kz} \hat{z}_t = \frac{1}{\beta} \hat{k}_{t-1} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) (V_{ck} \hat{k}_{t-1} + V_{cz} \hat{z}_t) + \frac{\tilde{\delta}}{\alpha\beta} \hat{z}_t.$$

Collecting coefficients:

$$\left[\frac{1}{\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) V_{ck} - V_{kk} \right] \hat{k}_{t-1} + \left[\frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) V_{cz} - V_{kz} \right] \hat{z}_t = 0.$$

Hence

$$\boxed{V_{kk} = \frac{1}{\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) V_{ck}} \quad (1)$$

and

$$\boxed{V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) V_{cz}} \quad (2)$$

3) From

$$\sigma E_t \hat{c}_{t+1} + \tilde{\delta}(1 - \alpha) \hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1} = \sigma \hat{c}_t,$$

substitute

$$\hat{c}_{t+1} = V_{ck} \hat{k}_t + V_{cz} \hat{z}_{t+1}, \quad \hat{k}_t = V_{kk} \hat{k}_{t-1} + V_{kz} \hat{z}_t, \quad E_t \hat{z}_{t+1} = \psi \hat{z}_t.$$

Then

$$\sigma E_t [V_{ck} \hat{k}_t + V_{cz} \hat{z}_{t+1}] + \tilde{\delta}(1 - \alpha) \hat{k}_t - \tilde{\delta} \psi \hat{z}_t = \sigma (V_{ck} \hat{k}_{t-1} + V_{cz} \hat{z}_t).$$

That is,

$$\sigma E_t [V_{ck} (V_{kk} \hat{k}_{t-1} + V_{kz} \hat{z}_t) + V_{cz} \hat{z}_{t+1}] + \tilde{\delta}(1 - \alpha) (V_{kk} \hat{k}_{t-1} + V_{kz} \hat{z}_t) - \tilde{\delta} \psi \hat{z}_t = \sigma V_{ck} \hat{k}_{t-1} + \sigma V_{cz} \hat{z}_t.$$

Collecting coefficients:

$$\left[\sigma V_{ck} V_{kk} + \tilde{\delta}(1 - \alpha) V_{kk} - \sigma V_{ck} \right] \hat{k}_{t-1} + \left[\sigma (V_{ck} V_{kz} + V_{cz} \psi) + \tilde{\delta}(1 - \alpha) V_{kz} - \tilde{\delta} \psi - \sigma V_{cz} \right] \hat{z}_t = 0.$$

Hence

$$\boxed{\sigma V_{ck} (1 - V_{kk}) = \tilde{\delta}(1 - \alpha) V_{kk}} \quad (3)$$

and

$$\boxed{\sigma V_{cz} (1 - \psi) = [\sigma V_{ck} + \tilde{\delta}(1 - \alpha)] V_{kz} - \tilde{\delta} \psi} \quad (4)$$

Solve the 4 Equations for 4 Unknowns

Collect the four equations and solve for the four unknowns.

(i) (1) and (3) are two equations in (V_{kk}, V_{ck}) . Eliminate V_{ck} .

From (1),

$$V_{ck} = \frac{\frac{1}{\beta} - V_{kk}}{\frac{\tilde{\delta}}{\alpha\beta} - \delta}.$$

Substitute into (3):

$$\sigma \left(\frac{\frac{1}{\beta} - V_{kk}}{\frac{\tilde{\delta}}{\alpha\beta} - \delta} \right) (1 - V_{kk}) = \tilde{\delta}(1 - \alpha)V_{kk}.$$

So

$$\frac{\sigma}{\beta} - \frac{\sigma}{\beta}V_{kk} - \sigma V_{kk} + \sigma V_{kk}^2 = \tilde{\delta}(1 - \alpha) \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) V_{kk}.$$

Equivalently,

$$V_{kk}^2 - \left[1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1 - \alpha)}{\sigma} \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) \right] V_{kk} + \frac{1}{\beta} = 0.$$

Let

$$\gamma = 1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1 - \alpha)}{\sigma} \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right).$$

Then the two roots are

$$V_{kk,1} = \frac{\gamma - \sqrt{\gamma^2 - \frac{4}{\beta}}}{2} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} > 0,$$

$$V_{kk,2} = \frac{\gamma + \sqrt{\gamma^2 - \frac{4}{\beta}}}{2} = \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} > 0.$$

We need to choose the stable root of the linear rational expectations (LRE) system.

Since

$$\gamma = 1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1 - \alpha)}{\sigma} \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) > 1 + \frac{1}{\beta},$$

we have

$$(1 - V_{kk,1})(1 - V_{kk,2}) = 1 + V_{kk,1}V_{kk,2} - (V_{kk,1} + V_{kk,2}) = 1 + \frac{1}{\beta} - \gamma < 0.$$

Hence

$$0 < V_{kk,1} < 1 < V_{kk,2}.$$

So $V_{kk,2}$ is the explosive root and should be discarded.

To see this, consider the shutdown-shock case $\hat{z}_t = 0$:

$$\hat{k}_t = V_{kk} \hat{k}_{t-1}.$$

Then V_{kk} is the speed of convergence:

$$|V_{kk}| < 1 \Rightarrow \text{deviation in capital shrinks over time, economy returns to S.S.,}$$

$$|V_{kk}| > 1 \Rightarrow \text{deviation in capital grows over time, economy explodes and violates TVC.}$$

By the Blanchard–Kahn condition, there is one predetermined state $\hat{k}_{t-1}$ and one jump variable $\hat{c}_t$, so the number of stable roots must equal the number of predetermined states. Therefore, exactly one stable root is needed.

Drop $V_{kk,2}$; only $V_{kk,1}$ is admissible.

Thus

$$V_{kk} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}.$$

And from (1),

$$V_{ck} = \frac{1}{\frac{\tilde{\delta}}{\alpha\beta} - \delta} \left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} \right).$$

(ii) (2) and (4) are two equations in (V_{kz}, V_{cz}) . Solve for V_{cz} by substituting V_{kz} out.

Starting from

$$\sigma V_{cz}(1 - \psi) = [\sigma V_{ck} + \tilde{\delta}(1 - \alpha)] V_{kz} - \tilde{\delta}\psi,$$

and

$$V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) V_{cz},$$

substitute the second into the first:

$$\begin{aligned} \sigma V_{cz}(1 - \psi) &= \left[\sigma \frac{1}{\frac{\tilde{\delta}}{\alpha\beta} - \delta} \left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} \right) + \tilde{\delta}(1 - \alpha) \right] \left[\frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) V_{cz} \right] - \tilde{\delta}\psi \\ \Rightarrow V_{cz} &= \frac{\sigma \frac{1}{\frac{\tilde{\delta}}{\alpha\beta} - \delta} \left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} \right) + \tilde{\delta}(1 - \alpha) - \alpha\beta\psi}{\sigma(1 - \psi) + \left[\sigma \frac{1}{\frac{\tilde{\delta}}{\alpha\beta} - \delta} \left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} \right) + \tilde{\delta}(1 - \alpha) \right] \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right)} \times \frac{\tilde{\delta}}{\alpha\beta}. \end{aligned}$$

$$\Rightarrow V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \frac{\tilde{\delta}}{\alpha\beta} \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) \cdot \frac{\sigma \frac{1}{\tilde{\delta}} \left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} \right) + \tilde{\delta}(1 - \alpha) - \alpha\beta\psi}{\sigma(1 - \psi) + \left[\sigma \frac{1}{\tilde{\delta}} \left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} \right) + \tilde{\delta}(1 - \alpha) \right] \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right)}.$$

Now we have:

$$\begin{cases} \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \\ \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t, \\ \hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t. \end{cases}$$

Where,

$$\begin{aligned} V_{kk} &= \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}, & V_{ck} &= \frac{\bar{K}}{\bar{C}} \left(\frac{1}{\beta} - V_{kk} \right), \\ V_{cz} &= \frac{\sigma V_{ck} + \tilde{\delta}(1 - \alpha) - \alpha\beta\psi}{\sigma(1 - \psi) + [\sigma V_{ck} + \tilde{\delta}(1 - \alpha)] \frac{\bar{C}}{\bar{K}}} \times \frac{\tilde{\delta}}{\alpha\beta}, \\ V_{kz} &= \frac{\tilde{\delta}}{\alpha\beta} - \frac{\bar{C}}{\bar{K}} V_{cz}. \end{aligned}$$

Feed in quarterly parameter values

$\beta = 0.99$ matches a 4.1% annual rate: $r_q = \frac{1}{\beta} - 1 \approx 1.01\%$,
 $r_a \approx (1 + r_q)^4 - 1 \approx 4.1\%$.

$\alpha = 0.36$ matches the capital income share in the data.

$\sigma = 1$ log utility. Usually $\sigma = 2$ implies $\text{IES} = \frac{1}{\sigma} = 0.5$.

$\delta = 0.025$ capital depreciates by 2.5% per quarter (about 10% annually).

$\bar{Z} = 1$ steady-state technology level is normalized.

$\psi = 0.95$ persistence of the productivity shock, i.e. how long ε_t lasts.

Then we can compute $(V_{kk}, V_{cz}, V_{kz}, V_{ck})$.

Impulse response function (IRF) Shock ε_t and trace how variables respond.

e.g. Trace out all variables for $\varepsilon_1 = 1$, $\varepsilon_t = 0$ for $t \geq 2$, start from S.S. (one-time temporary).

IRF and Half-life

Since

$$\hat{z}_t = \psi \hat{z}_{t-1} + \varepsilon_t,$$

we have

$$\hat{z}_1 = \psi \hat{z}_0 + \varepsilon_1 = \varepsilon_1,$$

since we start from steady state and $\hat{z}_0 = 0$. (Recall a hat means deviation from steady state.)

Also,

$$\hat{z}_2 = \psi \hat{z}_1 + \varepsilon_2 = \psi \hat{z}_1 = \psi \varepsilon_1,$$

and in general

$$\hat{z}_t = \psi^{t-1} \varepsilon_1.$$

When $|\psi| < 1$, the shock dies out as $t \rightarrow \infty$.

For capital,

$$\hat{k}_1 = V_{kk} \hat{k}_0 + V_{kz} \hat{z}_1 = V_{kz} \varepsilon_1,$$

since $\hat{k}_0 = 0$ at steady state.

Then

$$\hat{k}_2 = V_{kk} \hat{k}_1 + V_{kz} \hat{z}_2 = V_{kk} V_{kz} \varepsilon_1 + V_{kz} \psi \varepsilon_1 = (V_{kk} + \psi) V_{kz} \varepsilon_1.$$

In general,

$$\hat{k}_t = V_{kk} \hat{k}_{t-1} + V_{kz} \psi^{t-1} \varepsilon_1 = \sum_{i=0}^{t-1} V_{kk}^i \psi^{t-i-1} V_{kz} \varepsilon_1.$$

V_{kk}^i and V_{kz} are constant. Each past TFP effect ψ^{t-i-1} is stored/propagated through capital at rate V_{kk}^i , and V_{kz} translates the TFP shock into capital.

How to measure speed of convergence? Use half-life. We have

$$\hat{k}_{t+h} = V_{kk}^h \hat{k}_t.$$

The half-life is defined by

$$\frac{\hat{k}_{t+h_{1/2}}}{\hat{k}_t} = \frac{1}{2} \quad \Rightarrow \quad V_{kk}^{h_{1/2}} = \frac{1}{2} \quad \Rightarrow \quad \boxed{h_{1/2}^{(k)} = \frac{\ln(1/2)}{\ln(V_{kk})}} \quad (0 < V_{kk} < 1).$$

For the TFP shock,

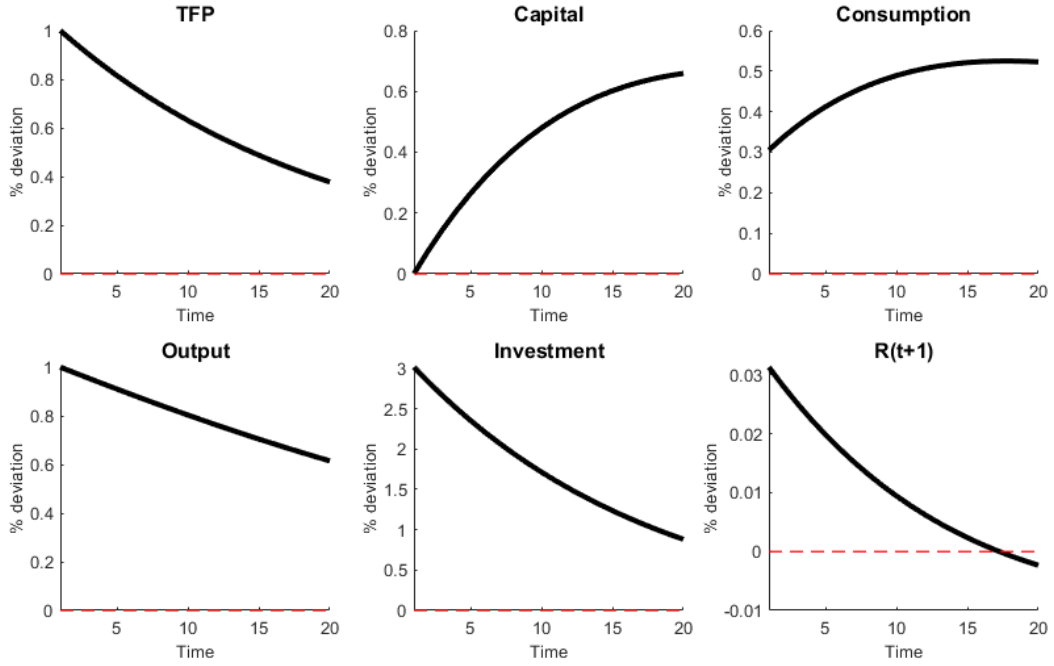
$$\boxed{h_{1/2}^{(z)} = \frac{\ln(1/2)}{\ln(\psi)}}.$$

Examples: if $V_{kk} = 0.95$,

$$h_{1/2}^{(k)} \approx 13.5 \text{ quarters} \approx 3.4 \text{ years}.$$

If $V_{kk} = 0.9$,

$$h_{1/2}^{(k)} \approx 6.6 \text{ quarters} \approx 1.6 \text{ years}.$$



Question: why is capital hump-shaped? From

$$\hat{k}_t = \sum_{i=0}^{t-1} (V_{kk}^i \psi^{t-i-1}) V_{kz} \varepsilon_1 = \begin{cases} V_{kz} \varepsilon_1 \frac{\psi^t - V_{kk}^t}{\psi - V_{kk}}, & \psi \neq V_{kk}, \\ V_{kz} \varepsilon_1 t \psi^t, & \psi = V_{kk}. \end{cases}$$

if $\psi = V_{kk}$, then

$$V_{kk}^i \psi^{t-i-1} = \psi^{t-1},$$

so

$$\hat{k}_t = \sum_{i=0}^{t-1} \psi^{t-1} V_{kz} \varepsilon_1 = t \psi^{t-1} V_{kz} \varepsilon_1.$$

Hence

$$\frac{\hat{k}_{t+1}}{\hat{k}_t} = \frac{(t+1)\psi^t}{t\psi^{t-1}} = \psi \left(1 + \frac{1}{t}\right).$$

For small t , $1 + \frac{1}{t}$ is large, so the ratio exceeds 1 and $\hat{k}_t$ keeps rising.

For large t , $1 + \frac{1}{t} \rightarrow 1$, so the ratio tends to $\psi < 1$ and $\hat{k}_t$ falls.

Peak occurs approximately when

$$\psi \left(1 + \frac{1}{t}\right) = 1 \quad \Rightarrow \quad t \approx \frac{\psi}{1 - \psi}.$$

Thus when $\psi = V_{kk}$, $\hat{k}_t$ rises at first, peaks around

$$t \approx \frac{\psi}{1 - \psi},$$

and then decays.

When $\psi \neq V_{kk}$,

$$\frac{\hat{k}_{t+1}}{\hat{k}_t} = \frac{\psi^{t+1} - V_{kk}^{t+1}}{\psi^t - V_{kk}^t} \Rightarrow \frac{\hat{k}_2}{\hat{k}_1} = \psi + V_{kk}.$$

We get an initial rise if $\psi + V_{kk} > 1$.

It peaks when $\hat{k}_{t+1} = \hat{k}_t$, i.e.

$$\psi^{t+1} - V_{kk}^{t+1} = \psi^t - V_{kk}^t,$$

so

$$\psi^t(1 - \psi) = V_{kk}^t(1 - V_{kk}),$$

hence

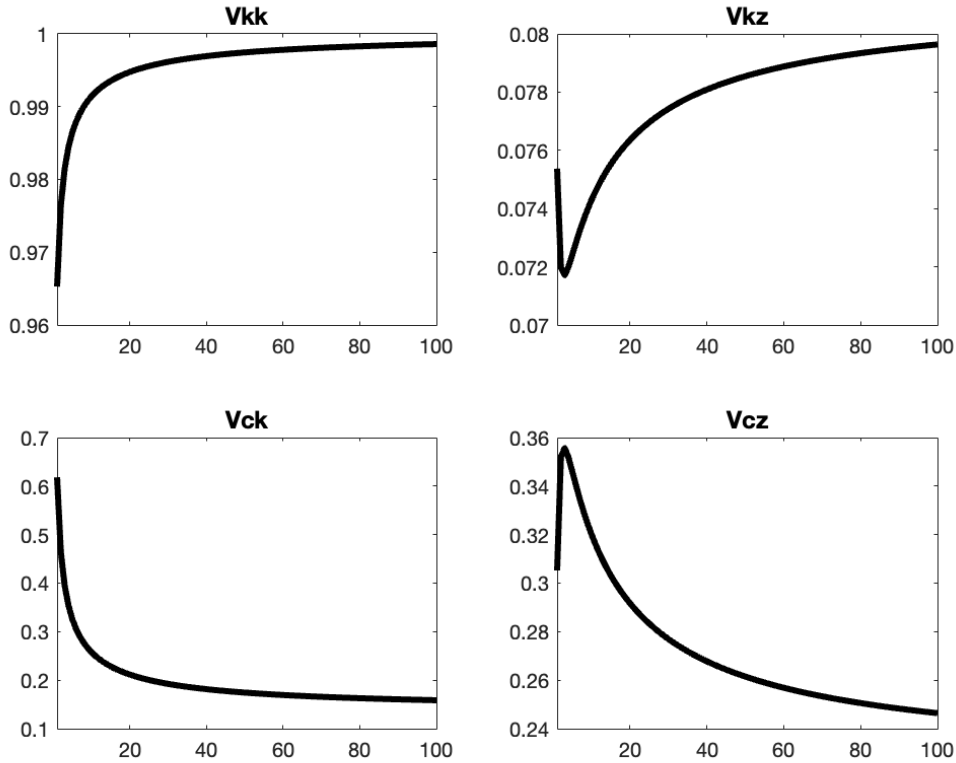
$$t^* = \frac{\ln\left(\frac{1-V_{kk}}{1-\psi}\right)}{\ln\left(\frac{\psi}{V_{kk}}\right)}.$$

If $t^* > 1$, the IRF rises for a while and then falls (hump-shaped). If $t^* \leq 1$, it peaks in the first period and then decays monotonically.

The effect of σ

Supply only depends on patience β and δ , labor is fixed.
 Demand: investment covers depreciation.
 $\left. \begin{array}{l} \text{Supply only depends on patience } \beta \text{ and } \delta, \text{ labor is fixed.} \\ \text{Demand: investment covers depreciation.} \end{array} \right\} \sigma \text{ does not have long-run effect.}$

But does have short-run effects, governed by σ .



1.

$$\sigma \uparrow \Rightarrow V_{kk} \uparrow \rightarrow 1$$

Capital becomes more persistent and convergence is slower.

Households dislike shifting consumption across time more, i.e. lower IES.

2.

$$\sigma \uparrow \Rightarrow IES = \frac{1}{\sigma} \downarrow \Rightarrow V_{ck} \downarrow.$$

Marginal utility becomes more responsive to changes in consumption, so consumption reacts less for a given state incentive coming from k_{t-1} .

3.

$$\sigma \uparrow \Rightarrow IES \downarrow \Rightarrow V_{cz} \downarrow$$

Consumption's direct response to the shock shrinks.

4.

$$\sigma \uparrow \Rightarrow IES \downarrow \Rightarrow V_{cz} \downarrow \Rightarrow V_{kz} \uparrow$$

After a TFP shock, consumption is more reluctant to jump, so more resources show up as investment / capital accumulation.

Recall the steady state of the system

$$\bar{C} = \bar{Z}\bar{K}^\alpha + (1 - \delta)\bar{K} - \bar{K},$$

$$1 = \beta[\alpha\bar{Z}\bar{K}^{\alpha-1} + (1 - \delta)].$$

Thus

$$\bar{K} = \left(\frac{\alpha\bar{Z}}{\frac{1}{\beta} - 1 + \delta} \right)^{\frac{1}{1-\alpha}},$$

$$\bar{Y} = \bar{Z}\bar{K}^\alpha, \quad \bar{C} = \bar{Y} - \delta\bar{K},$$

and

$$\frac{\bar{C}}{\bar{K}} = \frac{\bar{Y}}{\bar{K}} - \delta = \bar{Z}\bar{K}^{\alpha-1} - \delta = \frac{\frac{1}{\beta} - 1 + (1 - \alpha)\delta}{\alpha}.$$

The effect of δ (depreciation) Example values in the notes:

$$\delta = 0.025 : \quad \bar{K} \approx 38, \quad \bar{Y} \approx 3.7, \quad \bar{C} \approx 2.75,$$

$$\delta = 0.1 : \quad \bar{K} \approx 6.4, \quad \bar{Y} \approx 1.95, \quad \bar{C} \approx 1.3.$$

So higher depreciation implies a lower steady-state level.

Also,

$$\delta \uparrow \Rightarrow V_{kk} \downarrow, \quad \delta \uparrow \Rightarrow V_{kz} \uparrow.$$

With less capital stock and higher MPK, return and output both respond more proportionally.

For V_{cz} , the sign is ambiguous. The notes list several channels:

- Channel A: $\delta \uparrow \Rightarrow \tilde{\delta} \uparrow$,
- Channel B: $\delta \uparrow \Rightarrow \frac{\tilde{\delta}}{\alpha\beta} - \delta \uparrow$,
- Channel C: $\delta \uparrow \Rightarrow V_{kk} \downarrow, V_{ck} \uparrow$.

The final sign depends on calibration.

Complete procedure for solving RBC

- 1) Model setup: social planner's problem vs. competitive market (isomorphic).
- 2) Use dynamic programming to derive the necessary conditions(solve the model).
- 3) Write down the definition of equilibrium, including allocations and equilibrium conditions.
- 4) Find the steady state.
- 5) Log-linearize the dynamic system and determine local dynamic property.
- 6) Use the method of undetermined coefficients to solve the law of motion:

$$k_t = V_{kk}k_{t-1} + V_{kz}z_t,$$

and then solve for the other variables such as c_t .

- 7) Calibrate and simulate the model.

State Space Approach: 2-D State Space / Saddle Path Analysis

Start from

$$\hat{k}_t = \frac{1}{\beta} \hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}} \hat{c}_t + \frac{\tilde{\delta}}{\alpha\beta} \hat{z}_t.$$

Let $\hat{z}_t = 0$. Then

$$\hat{k}_t = \frac{1}{\beta} \hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}} \hat{c}_t,$$

so

$$\hat{c}_t = \frac{\bar{K}}{\beta\bar{C}} \hat{k}_{t-1} - \frac{\bar{K}}{\bar{C}} \hat{k}_t.$$

Also, from

$$\sigma E_t \hat{c}_{t+1} + \tilde{\delta}(1 - \alpha) \hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1} = \sigma \hat{c}_t,$$

letting $\hat{z}_t = 0$ gives

$$\sigma(\hat{c}_t - \hat{c}_{t+1}) - [1 - \beta(1 - \delta)](1 - \alpha) \hat{k}_t = 0$$

Rearrange:

$$\left\{ \begin{array}{l} \hat{k}_t - \hat{k}_{t-1} = \left(\frac{1}{\beta} - 1 \right) \hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}} \hat{c}_t, \\ \hat{c}_{t+1} - \hat{c}_t = \frac{1}{\sigma} [1 - \beta(1 - \delta)] (1 - \alpha) \underbrace{\left(\frac{\bar{C}}{\bar{K}} \hat{c}_t - \frac{1}{\beta} \hat{k}_{t+1} \right)}_{=-\hat{k}_t} \end{array} \right.$$

Steady-state loci If

$$\hat{k}_t = \hat{k}_{t-1},$$

then

$$\hat{c}_t = \left(\frac{1}{\beta} - 1 \right) \frac{\bar{K}}{\bar{C}} \hat{k}_{t-1}. \quad (1)$$

If

$$\hat{c}_{t+1} = \hat{c}_t,$$

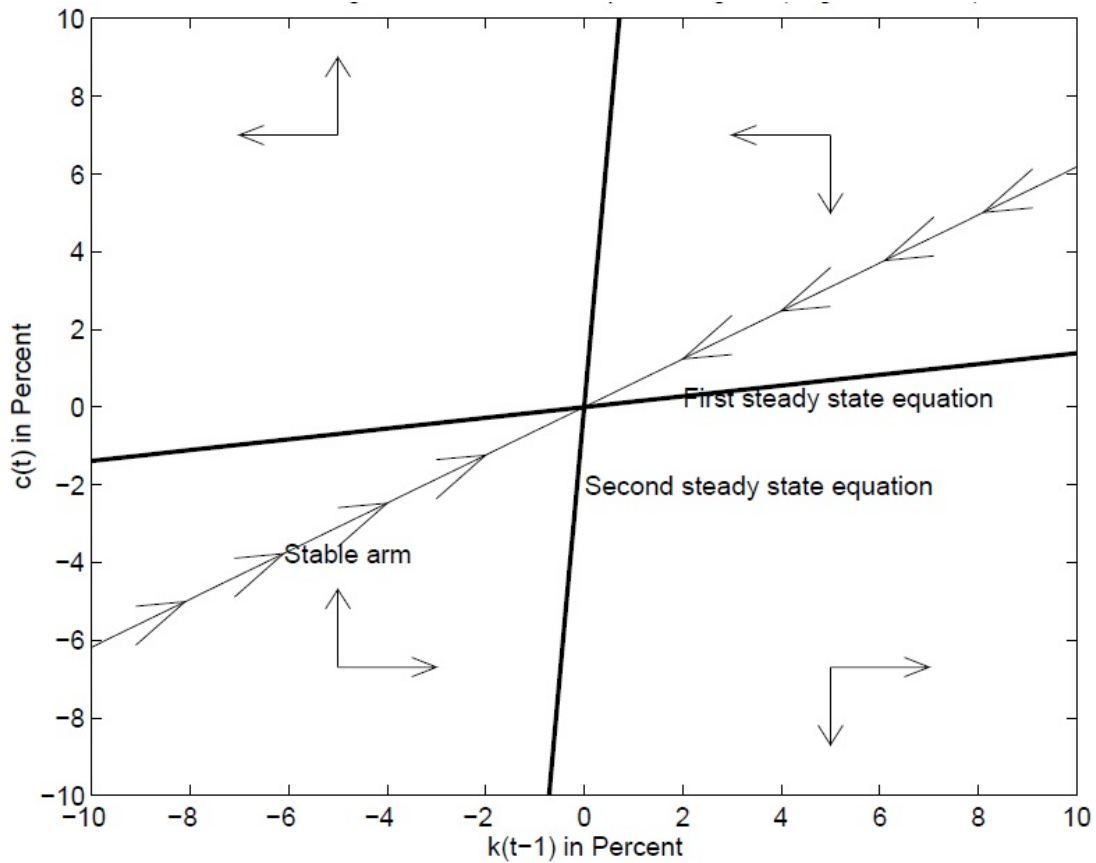
then

$$\hat{c}_t = \frac{\bar{K}}{\beta\bar{C}} \hat{k}_{t-1}. \quad (2)$$

The steady state is

$$\hat{k}_t = \hat{c}_t = 0.$$

The stable arm is the unique saddle path converging to the steady state.



The notes' directional interpretation:

- If $\hat{c}_t$ is above line (1), then $\hat{k}_t < \hat{k}_{t-1}$: consume too much, investment is low, so capital falls.
- If $\hat{c}_t$ is below line (1), then $\hat{k}_t > \hat{k}_{t-1}$: consume too little, save more, so capital rises.
- If $\hat{c}_t$ is above / left of line (2), then $\hat{c}_{t+1} > \hat{c}_t$: consume more next period.
- If $\hat{c}_t$ is below / right of line (2), then $\hat{c}_{t+1} < \hat{c}_t$: consume less next period.

Why These Relations Hold; Redo in Levels

Why line (1) implies $\hat{k}_t < \hat{k}_{t-1}$ If $\hat{c}_t$ is above line (1), then

$$\hat{c}_t > \left(\frac{1}{\beta} - 1\right) \frac{\bar{K}}{\bar{C}} \hat{k}_{t-1},$$

so

$$\frac{\bar{C}}{\bar{K}} \hat{c}_t - \left(\frac{1}{\beta} - 1\right) \hat{k}_{t-1} > 0.$$

Using

$$\hat{k}_t - \hat{k}_{t-1} = \left(\frac{1}{\beta} - 1\right) \hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}} \hat{c}_t,$$

we get

$$\hat{k}_t < \hat{k}_{t-1}.$$

Why line (2) implies $\hat{c}_{t+1} > \hat{c}_t$ If $\hat{c}_t$ is above line (2), then

$$\hat{c}_t > \frac{\bar{K}}{\beta \bar{C}} \hat{k}_{t+1},$$

so

$$\frac{\bar{C}}{\bar{K}} \hat{c}_t - \frac{1}{\beta} \hat{k}_{t+1} > 0.$$

Hence, from

$$\hat{c}_{t+1} - \hat{c}_t = \frac{1}{\sigma} [1 - \beta(1 - \delta)] (1 - \alpha) \left(\frac{\bar{C}}{\bar{K}} \hat{c}_t - \frac{1}{\beta} \hat{k}_{t+1} \right),$$

we have

$$\hat{c}_{t+1} > \hat{c}_t.$$

Stable arm The stable arm is the unique saddle path converging to steady state:

$$\hat{c}_t = V_{ck} \hat{k}_{t-1}.$$

The speed of convergence is

$$\hat{k}_t = V_{kk} \hat{k}_{t-1} \Rightarrow \hat{k}_{t+h} = V_{kk}^h \hat{k}_t.$$

Nature gives $\hat{k}_{t-1}$, and households choose $\hat{c}_t$ so as to land exactly on the stable arm. Then the economy follows that arm back to steady state. Off the stable arm, the system is unstable and diverges.

Redo in levels Set $Z_t = \bar{Z}$:

$$C_t = \bar{Z} K_{t-1}^\alpha + (1 - \delta) K_{t-1} - K_t,$$

$$1 = \beta E_t \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} (\alpha \bar{Z} K_t^{\alpha-1} + 1 - \delta).$$

So

$$K_t - K_{t-1} = \bar{Z} K_{t-1}^\alpha - \delta K_{t-1} - C_t,$$

and

$$E_t \frac{C_{t+1}}{C_t} = \left\{ \beta \left[\alpha \bar{Z} (\bar{Z} K_{t-1}^\alpha + (1 - \delta) K_{t-1} - C_t)^{\alpha-1} + 1 - \delta \right] \right\}^{1/\sigma}$$

At steady state, $\Delta K_t = 0$, so

$$C_t = \bar{Z} K_{t-1}^\alpha - \delta K_{t-1}. \tag{1}$$

Thus:

$$\text{if } C_t > \bar{Z} K_{t-1}^\alpha - \delta K_{t-1}, \quad K_t < K_{t-1},$$

$$\text{if } C_t < \bar{Z} K_{t-1}^\alpha - \delta K_{t-1}, \quad K_t > K_{t-1}.$$

At steady state, $\Delta C_t = 0$, so

$$1 = \left\{ \beta \left[\alpha \bar{Z} (\bar{Z} K_{t+1}^\alpha + (1 - \delta) K_{t+1} - C_t)^{\alpha-1} + 1 - \delta \right] \right\}^{1/\sigma},$$

which implies

$$C_t = \bar{Z} K_{t-1}^\alpha + (1 - \delta) K_{t-1} - \bar{K}. \quad (2)$$

Then:

$$\text{if } C_t > -\bar{K} + \bar{Z} K_{t-1}^\alpha + (1 - \delta) K_{t-1}, \quad C_{t+1} > C_t,$$

$$\text{if } C_t < -\bar{K} + \bar{Z} K_{t-1}^\alpha + (1 - \delta) K_{t-1}, \quad C_{t+1} < C_t.$$

Steady-state equations in levels

$$\begin{cases} C_t = \bar{Z} K_{t-1}^\alpha - \delta K_{t-1}, & (1) \\ C_t = \bar{Z} K_{t-1}^\alpha + (1 - \delta) K_{t-1} - \bar{K}. & (2) \end{cases}$$

Why is the stable arm still linear? Since

$$\hat{c}_t = V_{ck} \hat{k}_{t-1},$$

we have

$$\frac{C_t - \bar{C}}{\bar{C}} = V_{ck} \frac{K_t - \bar{K}}{\bar{K}}.$$

Thus

$$C_t = \frac{V_{ck} \bar{C}}{\bar{K}} K_t - V_{ck} \bar{C} + \bar{C}.$$

So in levels the stable arm is linear, with slope

$$\frac{V_{ck} \bar{C}}{\bar{K}}.$$

